

Chengtao Xu

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EDUCATION

Yale University

Postdoctoral Fellow, Molecular Biophysics and Biochemistry

New Haven, CT, United States

Jun.2024-Present

Cornell University

Postdoctoral Associate, Molecular Biology and Genetics

Ithaca, NY, United States

Mar. 2023-Jun.2024

Southeast University

Ph.D., Biomedical Engineering

Nanjing, China

Sep. 2018- Mar. 2023

Southeast University

B.S., Biomedical Engineering

Nanjing, China

Sep. 2014-Sep. 2018

RESEARCH EXPERIENCE

Yale University

New Haven, CT, United States

Cornell University

Ithaca, NY, United States

Postdoctoral Fellow, Supervisor: Dr. Ailong Ke

Mar. 2023-Present

- Converting IscB/Cas9 to a versatile RNA-guided RNA editor
- Structural insights to improve ssDNA-/RNA-targeting IscB/Cas9 in vivo performance
- Structural studies on CRISPR IVB/IVC complexes
- Structural studies on CRISPR I-E complexes for phage defense

Southeast University

Nanjing, China

Graduate Researcher; Advisor: Prof. Hong Liu

Jun. 2019- Mar. 2023

Versatile DNA-based data storage based on the principle of movable type printing

Integrated DNA data storage system based on electrochemical DNA synthesis and sequencing

Southeast University

Nanjing, China

Graduate Research Assistant; Advisor: Dr. Biao Ma

Sep. 2018- Jun. 2019

High-resolution patterning of liquid metal on hydrogels to construct flexible electronics

Southeast University

Nanjing, China

Undergraduate Research Program; Advisor: Prof. Hong Liu

Sep. 2017-Sep.2018

Ultrasensitive arsenic detection based on catalytic reduction of Rhodamine B

PEER-REVIEWED PUBLICATIONS

Chengtao Xu, Xiaolin Niu, Haifeng Sun, Hao Yan, Weixin Tang, Ailong Ke* Conversion of IscB and Cas9 into RNA-guided RNA editors. *Cell*, 2025, 188, 21, 5847-5861.

Chengtao Xu, Qi Yang, Xiaolin Niu, Ailong Ke* Structure basis for single-strand nucleic acid targeting by IscB and variants. *Nucleic Acids Research*, 2026, 54, 12, gkag607.

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Conor Pittman[#], **Chengtao Xu[#]**, Ryan J. Catchpole, Sandra Garrett, Ryan Fuchs, Xiaomeng Chu, Kira S. Makarova, Eugene V. Koonin, Peng Zhao, Lance Wells, Brenton R. Graveley, Ailong Ke*, Michael P. Terns* Type IV-C CRISPR-Cas effector complexes recognize double-stranded DNA and switch on collateral cleavage of ssDNA and RNA. *Cell reports*, in press.

Chengtao Xu, Biao Ma, Zhongli Gao, Xing Dong, Chao Zhao, Hong Liu* Electrochemical DNA synthesis and sequencing on a single electrode with scalability for integrated data storage. *Science Advances*, 2021, 7, eabk0100.

Chengtao Xu[†], Chao Zhao[†], Biao Ma and Hong Liu* Uncertainties in synthetic DNA-based data storage. *Nucleic Acids Res.*, 2021, 49, 10, 5451-5469. (Review)

Chengtao Xu, Biao Ma, Xing Dong, Lanjie Lei, Qing Hao, Chao Zhao,* Hong Liu,* Assembly of Reusable DNA Blocks for Data Storage Using the Principle of Movable Type Printing. *ACS Appl. Mater. Interfaces*, 20230, 15, 20, 24097-24108.

Chengtao Xu[†], Biao Ma[†], Shuai Yuan, Chao Zhao,* and Hong Liu* High-Resolution Patterning of Liquid Metal on Hydrogel for Flexible, Stretchable, and Self-Healing Electronics. *Adv. Electron. Mater.*, 2020, 6, 1900721.

Dezhi Feng, Biao Ma, **Chengtao Xu**, Jingyi Xu, Yiyang Zhang, Yi Chen, Gangsheng Chen, Yakun Gao, Duxin Chen, Wenwu Yu, Chao Zhao*, and Hong Liu*, Renewable electrode array based on transformable metal for scaling up DNA synthesis and manipulation. *Adv. Funct. Mater.*, 2025, 36, e13703

Yakun Gao, Biao Ma*, Gangsheng Chen, **Chengtao Xu**, Ziyang Kong, Yanjie Chen, Chao Zhao, Duxin Chen, Wenwu Yu, and Hong Liu*, Transformable and stimuli-responsive liquid metal for integrated, sustainable, and biomimetic DNA-based data storage. *Matter*, 2025, 8, 102145.

Biao Ma, **Chengtao Xu**, Lishan Cui, Chao Zhao* and Hong Liu* Magnetic Printing of Liquid Metal for Perceptive Soft Actuators with Embodied Intelligence. *ACS Appl. Mater. Inter.*, 2021, 13, 4, 5574-5582.

Biao Ma, **Chengtao Xu**, Junjie Chi, Jian Chen, Chao Zhao* and Hong Liu,* A Versatile Approach for Direct Patterning of Liquid Metal Using Magnetic Field. *Adv. Funct. Mat.*, 2019, 29, 1901370.

H-index: 13 (as of September 8, 2026)

(Other publications please see: <https://scholar.google.com/citations?user=6VnVUPcAAAAJ&hl=en>)

GRANTS & AWARDS

Tom and Joan Steitz RNA Fellowship award	Apr. 2026
Tom and Joan Steitz RNA Fellows program 2026 class, mentor: Ailong Ke, Yale University	

Poster Award	Jun. 2026
CRISPR 2026 International conference	

Poster Award	Oct. 2025
2025 Annual Retreat of the Yale Center for RNA Science and Medicine	

National Scholarship	Nov. 2021
Highest scholarship given by Chinese government (Top 0.2%)	

Excellent Bachelor Thesis Award	Sep. 2018
Southeast University and Jiangsu province (Top 1%)	

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Dec. 2017/Nov. 2015

Presidential Scholarship

Southeast University (Top 1%)

Chien-shiung Wu Scholarship for Biomedical Engineering

Jun. 2015

Southeast University (Top 2%)

First Prize, National Innovation and Design Competition of Biomedical Engineering

Nov. 2016

Designed a portable detection platform based on a microfluidic paper chip (Top 3%)

POSTERS & PRESENTATIONS

Conor Pittman[#], **Chengtao Xu[#]**, Ailong Ke^{*}, and Mike Terns^{*} Type IV-C CRISPR-Cas effector complexes recognize double-stranded DNA and switch on collateral cleavage of ssDNA and RNA. Poster. CRISPR 2026 International conference. New Haven, United States, June 2026.

Chengtao Xu and Ailong Ke^{*} Conversion of IscB and Cas9 into RNA-guided RNA-editors. Poster. 2025 Annual Retreat of the Yale Center for RNA Science and Medicine. New Haven, United States, November 2025.

Chengtao Xu and Ailong Ke^{*} Conversion of IscB and Cas9 into RNA-guided RNA-editors. Poster. Cold Spring Harbor Laboratory Genome Engineering: CRISPR Frontiers. New York, United States, August 2025/2026.

Chengtao Xu and Hong Liu^{*} Integrated data storage system based on DNA electrochemical synthesis and sequencing. Oral presentation. Joint Forum on Bioelectronics and Biophotonics. Hainan, China, December 2021.

Chengtao Xu[†], Biao Ma[†], and Hong Liu^{*} High-Resolution Patterning of Liquid Metal on Hydrogel for Flexible, Stretchable, and Self-Healing Electronics. Online oral presentation. International Conference on Biomedical and Bioinformatics Engineering. N0028. Kyoto, Japan, November 2020.

Chengtao Xu[†], Biao Ma[†], and Hong Liu^{*} High-Resolution Patterning of Liquid Metal on Hydrogel for Flexible, Stretchable, and Self-Healing Electronics. Poster. 3rd International Conference on Flexible Electronics. 103. Hangzhou, China, July 2019.

Chengtao Xu and Hong Liu Ultrasensitive Point-of-Care Testing of Arsenic Based on Catalytic Reaction of Unmodified Gold Nanoparticles. Poster. Asian Conference on Nanoscience and Nanotechnology. Qingdao, China, October 2018.

LEADERSHIP & COMMUNITY SERVICE

Mentoring, Ailong Ke Group, Yale University / Cornell University

Mar. 2023-Present

- Mentored rotation graduate students in CRISPR biology and protein engineering, fostering scientific enthusiasm and rigorous research standards
- Mentored visiting undergraduate students at Cornell during summer research programs, supporting independent project completion

Mentoring, Hong Liu Group, Southeast University

Sep. 2018-Mar. 2023

- Mentored undergraduate and incoming graduate students through independent projects, from experimental design to manuscript writing; several projects contributed to co-authored publications (*Adv. Funct. Mater.* 2025; *Matter* 2025)

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- Trained new group members in electrochemical measurement, gel electrophoresis, and microfluidic device fabrication

Organizing Committee, CRISPR 2026 International Conference, New Haven

Jun. 2026

- Served on the organizing committee for an international meeting of around 110 participants; coordinated poster session logistics and on-site operations

Scientific Community Engagement, Yale University

Jun. 2024-Present

- Founded and organize the departmental Friday scientific communication sessions, a weekly cross-group forum in MB&B
- Present research regularly in the MB&B Research-in-Progress series and the Yale RNA Club
- Served as ad hoc reviewer with Dr. Ke for Nature, Nature Communications, etc.

Laboratory Assistant, Hong Liu Group, Southeast University

Dec. 2020-Present

- Managed laboratory functions including organization, ordering, laboratory safety management, and scheduling equipment use.
- Trained new graduate students with basic experimental techniques including electrochemical measurements, gel electrophoresis, microfluidic device fabrication.
- Mentored students on new research topics (e.g., using primer exchange reaction to amplify DNA aptamers for pathogen detection).